# EMERGENT GEOMETRY FROM COARSE-GRAINED OBSERVABLE ALGEBRAS: THE HOLOGRAPHIC ALAYA–FIELD FRAMEWORK

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**Preliminary Remark (Structural Stance).** In much of modern theoretical physics, it is tacitly assumed that the decomposition of a system into subsystems is either physically given or at least unproblematic. Hilbert spaces are factorized, degrees of freedom are labeled, and geometry is inferred from relations between these parts.

In this work, we adopt a different structural stance. We treat the universal quantum state as given, but regard subsystem structure, locality, and geometry as secondary constructs arising from restrictions on observable algebras. The guiding question is not how geometry emerges from quantum states, but how different effective realities can emerge from the *same* state once no preferred factorization is assumed.

This shift is modest in formalism but radical in implication: it relocates the origin of structure from states to algebras, and from kinematics to accessibility.

**ABSTRACT**

We construct a theoretical framework where the tensor factorization of a Hilbert space is treated as a structural variable rather than a kinematic background. By lifting the "Alaya" concept to a globally entangled vacuum state $|\Psi_{\text{vac}}\rangle$, we analyze how local geometric data may depend on specific observable subalgebras $\mathcal{A}_i \subset \mathcal{B}(\mathcal{H})$ when an explicit reconstruction dictionary is supplied. We show that, under a separating reconstruction dictionary, inequivalent entropy profiles induced by coarse-graining maps yield reconstructed geometric data not related by diffeomorphism. The analysis is structural in nature: we do not propose new dynamics, but examine consistency and consequences of removing subsystem factorization from fundamental assumptions. This approach is complementary to existing interpretational frameworks and suggests natural connections to algebraic quantum field theory, entanglement-based approaches to spacetime, and quantum information theory.

## 1 Introduction

### 1.1 Motivation

Two central problems in contemporary theoretical physics concern the emergence of classicality and the emergence of geometry. In quantum foundations, the measurement problem asks how effectively classical behavior arises from an underlying quantum description. Decoherence theory has provided a powerful account of this process by explaining the suppression of interference between certain degrees of freedom through environmental entanglement [1]. However, this explanation typically presupposes a fixed decomposition of the total system into subsystems, distinguishing "system," "apparatus," and "environment" from the outset.

A closely related emergence problem appears in quantum gravity. A growing body of work suggests that spacetime geometry is not fundamental, but arises from patterns of quantum entanglement [2, 3]. In holographic settings, geometric quantities are related to entanglement measures via precise correspondences, most notably the Ryu–Takayanagi

formula [4]. Yet these constructions likewise assume a prior specification of spatial regions or tensor factors, with geometry inferred only after such a subdivision has been fixed.

In both contexts, the emergence problem is addressed only after a subsystem decomposition has been assumed. The structure responsible for classicality or geometry is therefore explained relative to a partition whose origin remains largely unexamined.

### 1.2 The Structural Gap

The assumption of a given subsystem structure is often treated as innocuous, or as a matter of convenient description. However, from a fundamental perspective, there is no canonical tensor factorization of a generic Hilbert space, nor a unique way to decompose a global quantum state into subsystems. While this issue is occasionally acknowledged in passing [6, 7], its consequences for emergence are rarely explored systematically.

The present work does not challenge the empirical success of decoherence theory, entanglement-based approaches to geometry, or the algebraic formulation of quantum theory. Rather, we make explicit a structural assumption common to these frameworks and investigate the consequences of relaxing it. Our focus is on what follows if subsystem structure itself is treated as emergent, rather than fundamental.

### 1.3 Our Contribution

We formulate a framework in which subsystem structure is not assumed *a priori*, but arises from a choice of coarse-graining over observable algebras. Within this framework, we show that different coarse-graining channels can induce inequivalent effective states and connectivity data. In a holographic or other reconstruction model that separates the resulting entropy profiles modulo diffeomorphisms, this difference can induce inequivalent effective geometries. Outside such a dictionary, the output is a weighted connectivity structure, not a spacetime metric.

The analysis is structural in nature. Our aim is not to propose a new dynamical mechanism, but to examine the consistency and consequences of removing subsystem factorization from the set of fundamental assumptions.

### 1.4 Terminology: The Alaya-Field

Throughout this work, we adopt the term *Alaya-Field* to denote the fundamental, non-factorized structure prior to any subsystem decomposition. This terminology, borrowed from Yogācāra philosophy (referring to the "storehouse consciousness"), is used here strictly in a technical sense.

In what follows, the Alaya-Field does not refer merely to a vector in Hilbert space, but to the triple

$$(\mathcal{H}_U, \mathcal{A}_U, |\Omega\rangle),$$

where $\mathcal{A}_U$ is a von Neumann algebra acting on $\mathcal{H}_U$ and $|\Omega\rangle$ is a cyclic and separating vector. The emphasis is on the absence of a canonical factorization, not on the particular choice of state.

In an AQFT realization, $\mathcal{A}_U$ may be a Type $\mathrm{III}_1$ factor and $|\Omega\rangle$ a cyclic and separating vector [8]. This is a possible source-side regime, not a consequence of the finite-dimensional calculations below, and we do not infer localized eigenmodes or a metric from the Type III label alone. This usage is intended to evoke the non-factorized, pre-geometric nature of the fundamental quantum structure, without importing any metaphysical commitments.

### 1.5 Regime of Each Result

The paper uses a mixed regime. The source triple $(\mathcal{H}_U, \mathcal{A}_U, |\Omega\rangle)$ may be discussed for a general von Neumann algebra, including Type III under the stated cyclic-and-separating hypothesis. The factorization, Haar measure, density-matrix, partial-trace, and von Neumann-entropy results in Sections 2–4 are finite-dimensional/Type I results. Type III factors have no density matrices or trace entropy, so those results are not asserted there. The conversion from entropy/connectivity data to a spacetime metric is a separate model-dependent holographic input.

### 1.6 Structure of the Paper

The paper is organized as follows. In Section 2, we review the absence of a canonical subsystem decomposition in quantum theory and formalize this observation. Section 3 introduces coarse-graining in terms of observable algebras and analyzes how effective subsystem descriptions arise from this procedure. In Section 4, we construct coarse-graining-dependent connectivity data, give a state-space QFIM construction, and state the additional reconstruction

hypothesis required before metric inequivalence can be claimed. Section 5 discusses the scope and conceptual implications of the framework, its relation to existing approaches, and possible directions for future work. We conclude in Section 6 with a summary of results and open questions.

## 2 Absence of Canonical Tensor Factorization

### 2.1 The Factorization Problem

In standard quantum mechanics, the state space of a composite system is constructed as a tensor product of subsystem Hilbert spaces. This construction presupposes that a natural decomposition into subsystems has already been identified. However, for a given Hilbert space $\mathcal{H}_{\text{total}}$, there is no unique or canonical way to express it as a tensor product $\mathcal{H}_A \otimes \mathcal{H}_B$ without additional physical input.

The universal entanglement statement used in an earlier version is withdrawn. Indeed, fix dimensions $d_A, d_B > 1$ with $d_A d_B = \dim \mathcal{H}_{\text{total}}$. For any pure state $|\Psi_U\rangle$, choose a unitary $U$ sending a reference product vector $|0\rangle_A \otimes |0\rangle_B$ to $|\Psi_U\rangle$. Pulling the reference tensor factors through $U$ gives a nontrivial factorization in which $|\Psi_U\rangle$ is a product. Thus no finite-dimensional pure state is entangled across *every* nontrivial factorization.

What does survive is a fixed-family typicality result.

**Theorem 2.1** (Entanglement typicality for a specified family)**.** *Let $\mathcal{H}_{\text{total}}$ be finite dimensional and let $\mathfrak{F} = \{\mathbf{f}_1, \ldots, \mathbf{f}_m\}$ be a finite family of nontrivial tensor factorizations selected by a physical rule independent of the sampled state. Write $\mathcal{H}_{\text{total}} = \mathcal{H}_{A_j} \otimes \mathcal{H}_{B_j}$ with $d_{A_j} \le d_{B_j}$. For $\delta > 0$, let*

$$p_j(\delta) := \Pr_{\Psi \sim \text{Haar}} \left[ S(\rho_{A_j}) < \log d_{A_j} - \delta \right].$$

*Then*

$$\Pr\left[ S(\rho_{A_j}) \ge \log d_{A_j} - \delta \text{ for every } j \right] \ge 1 - \sum_{j=1}^{m} p_j(\delta),$$

*and the standard concentration estimates make each $p_j(\delta)$ small at large dimension under the usual dimension-ratio hypotheses.*

*Proof.* For each fixed $\mathbf{f}_j$, Page's theorem gives [9]

$$\mathbb{E}[S(\rho_{A_j})] = \sum_{k=d_{B_j}+1}^{d_{A_j} d_{B_j}} \frac{1}{k} - \frac{d_{A_j} - 1}{2 d_{B_j}}, \tag{1}$$

while concentration of measure bounds the lower tail [12]; this is precisely $p_j(\delta)$. Applying the union bound to the finite, preselected family proves the displayed inequality. □

**Remark 2.2** (What the theorem does not select)**.** *Theorem 2.1 says nothing about all possible factorizations and supplies no preferred member of $\mathfrak{F}$. Specifying the admissible family and its selection rule is a quantum-mereology input, not a consequence of Haar typicality. Developing that input is an open problem; see [16, 17].*

**Remark 2.3** (Relation to prior work)**.** *The observation that tensor product structures are not canonical was recognized early in the foundations of quantum theory, and has been systematically analyzed in work on virtual subsystems and quantum mereology [5, 6, 16, 17]. Their work demonstrated that subsystem decompositions are effectively determined by sets of accessible observables rather than being intrinsic to the Hilbert space. The present framework extends this perspective to the context of holographic geometry and entanglement-based spacetime emergence, where the choice of observable algebra determines not only subsystem structure but also the emergent geometric description.*

**Remark 2.4.** *We do not claim that subsystem decompositions are impossible or unphysical, but that they are not uniquely determined by the universal quantum state alone. This observation motivates the search for additional structure that specifies how a subsystem decomposition arises in concrete physical contexts.*

### 2.2 Relation to Algebraic Approaches

The absence of a canonical factorization has been recognized in various contexts, including algebraic quantum field theory where observable algebras take precedence over tensor product structures [8]. Our framework builds on these insights by treating coarse-graining structure as the fundamental input from which subsystem decompositions emerge.

# 3 Observer-Dependent Coarse-Graining and Effective States

## 3.1 Coarse-Graining Structure

Before introducing the formal definition, we emphasize that the introduction of an observable algebra does not reinstate a tensor factorization. Operators may act irreducibly on $\mathcal{H}_{\text{total}}$ without inducing any subsystem decomposition. In algebraic quantum field theory, observable algebras are defined independently of any global tensor product structure [8]. Furthermore, a coarse-graining is not selected by an agent, but instantiated by a concrete physical interaction structure.

**Definition 3.1** (Operational Coarse-Graining Structure)**.** *Let $\mathcal{H}_U$ be the universal Hilbert space. A **coarse-graining structure** $\mathbf{c}$ is defined as a pair*

$$\boxed{\mathbf{c} \equiv (\mathcal{A}_{\text{sel},\mathbf{c}}, \Phi_{\mathbf{c}})}$$

*where:*

1. *$\mathcal{A}_{\text{sel},\mathbf{c}} \subset \mathcal{B}(\mathcal{H}_U)$ is a selected unital $*$-subalgebra used to define the effective representation. It is not identified with Paper E's probe-reachable effect system, pointer/record algebra, or environmental fixed set; a physical model must state any map between them.*
2. *In the finite-dimensional results below, $\Phi_{\mathbf{c}}$ is a CPTP map on states implementing an operational reduction, and its Heisenberg dual $\Phi_{\mathbf{c}}^*$ is unital and completely positive. Examples include partial trace over a specified factor and channels induced by a conditional expectation. In a general von Neumann setting the corresponding object would be a normal channel on states with a normal UCP dual, but no entropy theorem below is asserted merely from that extension.*

**Remark 3.2** (Non-uniqueness of observable algebras)**.** *It is important to note that the specification of a selected algebra $\mathcal{A}_{\text{sel},\mathbf{c}}$ is not assumed to be unique. In algebraic quantum field theory and quantum information theory, there exists no theorem guaranteeing a unique maximal observable algebra associated with a given physical system without additional structure. The coexistence of multiple admissible algebras reflects physical underdetermination rather than subjectivity or observer dependence.*

## 3.2 Refinement Structure of Coarse-Grainings

**Definition 3.3** (Refinement Relation)**.** *Given two coarse-graining structures $\mathbf{c}_1, \mathbf{c}_2$, we say $\mathbf{c}_1 \succeq \mathbf{c}_2$ if there exists a CPTP map*

$$\Lambda : \mathcal{B}(\mathcal{H}_{\mathit{eff}}(\mathbf{c}_1)) \to \mathcal{B}(\mathcal{H}_{\mathit{eff}}(\mathbf{c}_2))$$

*such that*

$$\Phi_{\mathbf{c}_2} = \Lambda \circ \Phi_{\mathbf{c}_1}$$

*In this case, $\mathbf{c}_2$ is a **further coarse-graining** of $\mathbf{c}_1$.*

**Remark 3.4** (Multiplicity of coarse-graining maps)**.** *For a fixed selected algebra $\mathcal{A}_{\text{sel},\mathbf{c}}$, there generally exist multiple completely positive trace-preserving maps implementing distinct coarse-graining procedures. The present framework does not require the selection of a preferred CPTP map. Rather, different maps correspond to physically realizable information-loss mechanisms, such as tracing over inaccessible degrees of freedom or effective decoherence channels.*

Different coarse-graining choices are related not by unitary symmetry, but by information-theoretic refinement maps, forming a partially ordered structure rather than a group.

## 3.3 Core Theorem

The absence of a unitary intertwiner and the distinction of observable images were previously presented as conclusions, although they merely unpacked (and in the case of the images did not follow from) the chosen inequivalence relation. They are not retained as theorem claims. The nontrivial finite-dimensional statement is the following generic separation result.

**Theorem 3.5** (Generic separation for a fixed pair of channels)**.** *Let $\mathcal{H}$ and $\mathcal{K} = \mathcal{K}_A \otimes \mathcal{K}_{A^c}$ be finite dimensional, and let $\Phi_1 \neq \Phi_2$ be two CPTP maps from states on $\mathcal{H}$ to states on the common output space $\mathcal{K}$. Then the pure inputs on which $\Phi_1$ and $\Phi_2$ agree form a Haar-null proper real-algebraic subset. If, in addition, the real-analytic function*

$$f_A(\Psi) := S(\mathrm{Tr}_{A^c}\, \Phi_1(|\Psi\rangle\langle\Psi|)) - S(\mathrm{Tr}_{A^c}\, \Phi_2(|\Psi\rangle\langle\Psi|))$$

*is not identically zero on each connected component under consideration of the open set where both reduced outputs are full rank, then $f_A(\Psi) \neq 0$ for generic inputs in those components.*

*Proof.* Because pure-state projectors span the Hermitian operators, $\Phi_1 - \Phi_2$ cannot vanish on every pure projector. The equality condition is polynomial in the real and imaginary components of $|\Psi\rangle$, so its solution set is a proper real-algebraic subset and has Haar measure zero. On the full-rank domain, von Neumann entropy and hence $f_A$ are real analytic. A nonzero real-analytic function has a measure-zero zero set, proving the conditional entropy statement. □

**Remark 3.6.** *The nondegeneracy hypothesis on $f_A$ is essential: distinct channels related by a unitary acting only on $A$ can produce different states with identical subsystem entropy for every input. The theorem therefore does not turn channel inequivalence alone into entanglement or geometric inequivalence.*

**Remark 3.7** (Relation to decoherence theory)**.** *We emphasize that the present framework is fully compatible with standard decoherence theory and does not modify its dynamical content. Decoherence successfully explains the emergence of classical behavior given a fixed system–environment decomposition. The novelty of the present approach lies instead in treating such subsystem decompositions as coarse-graining–dependent and not fundamental.*

### 3.4 Clarification on Subsystem Structure

Subsystems are not fundamental inputs to the framework, but derived representations induced by a chosen observable algebra. The connectivity measure defined in Section 4 acts on these representations, and is not used to define the algebra itself. The causal order is:

$$\text{Observable Algebra} \to \text{Representation} \to \text{Entanglement} \to \text{Connectivity} \to \text{Geometry}$$

No geometric structure is presupposed in the definition of coarse-graining.

### 3.5 Algebraic Perspective on Subsystems

A potential concern is whether the use of observable algebras implicitly presupposes a subsystem decomposition. We stress that this is not the case. Observable algebras need not be defined via a tensor factorization of the total Hilbert space.

In algebraic quantum field theory, local algebras are assigned to spacetime regions without invoking a global tensor product structure [8]. Subsystems arise only at the level of representations induced by a chosen algebra, rather than serving as its foundational input. In this sense, subsystem structure is emergent rather than fundamental.

**Remark 3.8** (Subsystems as induced representations)**.** *Within the present framework, subsystems are not primitive entities. They emerge as representations associated with a given observable algebra and coarse-graining structure. This avoids circularity by reversing the usual explanatory order: observable structure precedes subsystem identification.*

## 4 Entanglement Structure and Emergent Geometry

### 4.1 Mutual Information as Connectivity Measure

We stress that no geometric structure is assumed in the definition of the coarse-graining introduced in Section 3. Geometry only appears at the level of entanglement relations between the induced subsystems.

Given a coarse-graining structure $\mathbf{c} = (\mathcal{A}_{\mathrm{sel},\mathbf{c}}, \Phi_{\mathbf{c}})$ as defined in Section 3, the universal quantum state $|\Psi_U\rangle$ induces a family of effective subsystems $\{A, B, \dots\}$ associated with subalgebras of $\mathcal{A}_{\mathrm{sel},\mathbf{c}}$. For any such pair of subsystems $A$ and $B$, we consider the quantum mutual information

$$I_{\mathbf{c}}(A:B) = S_{\mathbf{c}}(A) + S_{\mathbf{c}}(B) - S_{\mathbf{c}}(AB), \tag{2}$$

where $S_{\mathbf{c}}(\cdot)$ denotes the von Neumann entropy computed after coarse-graining.

**Definition 4.1** (Entanglement-Induced Connectivity)**.** *Let $A$ and $B$ be two disjoint subsystems induced by* c*. We define an effective* ***proximity measure*** *$\mu_{\mathbf{c}}(A, B)$ based on the quantum mutual information:*

$$\mu_{\mathbf{c}}(A, B) := I_{\mathbf{c}}(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}). \tag{3}$$

*While $\mu_{\mathbf{c}}$ does not itself constitute a metric (as it violates the triangle inequality), it defines a weighted connectivity graph in a declared subsystem representation.*

Here is the precise finite-dimensional information-geometric construction used when a metric on a *parameter space of coarse-grained states* is needed. Let $\lambda \mapsto \rho_{\mathbf{c}}(\lambda)$ be a smooth faithful family and define the symmetric logarithmic derivatives $L_a$ by

$$\partial_a \rho_{\mathbf{c}} = \frac{1}{2}\big(L_a \rho_{\mathbf{c}} + \rho_{\mathbf{c}} L_a\big).$$

The SLD quantum Fisher metric is

$$g_{ab}^{\mathrm{QF}}(\lambda) := \frac{1}{2}\,\mathrm{Tr}[\rho_{\mathbf{c}}(\lambda)\{L_a, L_b\}]\,. \tag{4}$$

This is a positive-semidefinite Riemannian metric on identifiable parameter directions. It is not a Lorentzian spacetime metric, and it is not determined by the pairwise numbers $\mu_{\mathbf{c}}(A, B)$ without an additional map from graph/state parameters to geometry. In the holographic discussion below that additional map is a reconstruction dictionary, not a general consequence of QFIM [10].

This quantity should be understood as a *connectivity measure* rather than a fundamental spacetime distance. In general, mutual information does not define a metric on arbitrary quantum subsystems. However, it acquires a natural geometric interpretation under physically motivated assumptions, which we make explicit below.

**Assumptions.** Throughout this section, we restrict attention to coarse-grainings that satisfy a geometric admissibility condition:

**Definition 4.2** (Geometric Admissibility)**.** *A coarse-graining structure* $\mathbf{c}$ *is said to be* ***geometrically admissible*** *if the induced mutual information satisfies:*

1. Finite correlation length: *The state* $|\Psi_U\rangle$ *exhibits exponentially decaying correlations with respect to the induced subsystems, as is typical for gapped systems, ground states of local Hamiltonians, and holographic large-*$N$ *states.*
2. Monotonic decay under refinement: *Mutual information decreases monotonically as subsystems become more refined.*
3. Stability under perturbations: *The entanglement structure is robust under small perturbations of* $\Phi_{\mathbf{c}}$.

Not all coarse-grainings admit a geometric interpretation. Geometry is not generic; it is a *special phase* of information organization. We further assume coarse-graining consistency: all entropic quantities are computed using a single coarse-graining structure $\mathbf{c}$, avoiding any mixing of inequivalent observable algebras.

Under these assumptions, the entanglement-induced connectivity $\mu_{\mathbf{c}}(A, B)$ is non-negative, symmetric, and vanishes if and only if the subsystems are uncorrelated at the level resolved by $\mathbf{c}$. Moreover, in regimes where mutual information decays monotonically with separation, $\mu_{\mathbf{c}}$ induces an effective notion of spatial proximity.

**Remark 4.3.** *The connectivity* $\mu_{\mathbf{c}}(A, B)$ *should be understood as an* effective, coarse-graining–dependent measure of correlation*, rather than a fundamental spacetime metric. A metric on a state-family parameter space may be supplied by Eq.* (4)*. A spacetime metric requires the additional reconstruction input stated below.*

In a continuum limit of densely overlapping subsystems, the collection $\{\mu_{\mathbf{c}}(A, B)\}$ may approximate a weighted graph with a geometric interpretation. This is a model-building step: neither graph convergence nor a map from the QFIM parameter metric to spacetime is established in general.

### 4.2 From Entanglement to Geometry

The idea that spacetime geometry is encoded in quantum entanglement has been explored extensively in the context of holography. In particular, the Ryu–Takayanagi formula relates the entanglement entropy of a boundary region $A$ to the area of an extremal surface $\gamma_A$ in the bulk [4],

$$S(A) = \frac{\mathrm{Area}(\gamma_A)}{4G_N}, \tag{5}$$

suggesting a direct correspondence between entanglement structure and geometric data.

More generally, Van Raamsdonk has argued that the connectivity of spacetime is controlled by the pattern of entanglement between subsystems [3]. In this perspective, highly entangled degrees of freedom correspond to nearby regions in the emergent geometry, while weakly entangled subsystems are geometrically distant.

The entanglement-induced connectivity $\mu_{\mathbf{c}}(A, B)$ provides a concrete realization of this idea. Given a collection of subsystems induced by $\mathbf{c}$, the mutual information defines a weighted graph whose vertices correspond to subsystems and whose edge weights encode entanglement strength.

In tensor network constructions such as MERA, similar entanglement graphs admit a natural geometric interpretation, with graph distance approximating continuum spatial distance [11]. In particular tensor-network or holographic

models, an appropriate continuum limit can be equipped with an effective Riemannian spatial geometry whose metric reflects the underlying entanglement structure.

In this sense, geometry emerges not as an additional postulate, but as an effective description of how information is distributed and shared among coarse-grained degrees of freedom.

### 4.3 Coarse-Graining Dependence of Geometry

We now turn to the central observation of this section: the emergent geometry depends essentially on the choice of coarse-graining structure.

**Theorem 4.4** (Coarse-graining dependent geometry, conditional reconstruction)**.** *Let $\mathbf{c}_1$ and $\mathbf{c}_2$ act on the same finite-dimensional global state and induce entropy profiles $S_1, S_2$. Assume a specified reconstruction dictionary $\mathfrak{G}$ is defined on both profiles and is separating modulo diffeomorphisms on the compared class:*

$$\mathfrak{G}[S_1] \simeq_{\mathrm{diff}} \mathfrak{G}[S_2] \quad \Longrightarrow \quad S_1 \simeq_{\mathfrak{G}} S_2.$$

*If $S_1 \not\simeq_{\mathfrak{G}} S_2$, then the reconstructed metrics $\mathfrak{G}[S_1]$ and $\mathfrak{G}[S_2]$ are not related by a diffeomorphism. The theorem makes no metric claim when no such dictionary is supplied.*

*Proof.* This is the contrapositive of the separating property of $\mathfrak{G}$. Holographic entropy/area reconstruction is one setting in which such a property may be established on a restricted state class [4]. Channel inequivalence by itself does not provide either the hypothesis $S_1 \not\simeq_{\mathfrak{G}} S_2$ or the dictionary. □

**Example 4.5** (Free Fermion Chain: Spatial vs. Momentum Coarse-Graining)**.** *Consider $N$ free fermions on a one-dimensional lattice with ground state $|\Psi_0\rangle$.*

***Spatial coarse-graining** $\mathbf{c}_x$: group sites into blocks of $k$ consecutive sites. By the area law for gapped free-fermion chains [13], the entanglement entropy of a block of $L$ sites satisfies $S(L) \sim \mathrm{const}$, independent of $L$. The mutual information between blocks $A$ and $B$ separated by distance $r$ decays exponentially as $I(A:B) \sim e^{-r/\xi}$, where $\xi$ is the correlation length set by the spectral gap [14]. The resulting connectivity graph has short-range edges: its graph distance approximates a one-dimensional lattice, yielding a line-like emergent geometry.*

***Momentum-mode coarse-graining** $\mathbf{c}_p$: use the occupation-number tensor factors of the diagonal momentum modes. A free-fermion ground state is a Slater determinant*

$$|\Psi_0\rangle = \prod_{k \in F} c_k^\dagger |0\rangle,$$

*which is a product of filled or empty occupation states in this factorization. Every subset of momentum modes therefore has a pure reduced state and disjoint momentum subsets have zero mutual information. The momentum connectivity graph is empty, not a volume-law, long-range graph. Wolf's result [15] concerns* spatial *entanglement scaling in fermionic systems and does not support momentum-mode entanglement.*

*Thus the example demonstrates dependence of connectivity on a specified factorization: the spatial representation can carry short-range edges, while the free momentum-occupation representation carries none. It does not exhibit two reconstructed geometries; indeed $\mathbf{c}_p$ fails the geometric-admissibility goal unless further, interacting correlations or a separate reconstruction prescription are supplied.*

A simple conceptual illustration is provided by contrasting spatial coarse-graining with momentum-space coarse-graining. In the former case, entanglement typically obeys an area law and supports a local, connected geometry. For the free Slater determinant, the latter has no momentum-mode entanglement at all; interacting models would be required to obtain a nontrivial momentum graph. Importantly, both descriptions arise from the same underlying state $|\Psi_U\rangle$.

Thus, the geometry inferred from entanglement is not an intrinsic property of the quantum state alone, but depends on how information is rendered accessible through coarse-graining.

### 4.4 Beyond Coordinate Choice

It is important to distinguish the dependence described above from an ordinary change of coordinates. A diffeomorphism acts within a fixed algebra of observables, preserving the underlying notion of what is measurable. By contrast, a change of coarse-graining alters the observable algebra itself, modifying which correlations are accessible and how subsystems are defined.

These are categorically distinct operations. Nevertheless, algebraic inequivalence alone does not prove that two reconstructed metrics fail to be diffeomorphic: a reconstruction dictionary may discard the algebraic differences. Non-diffeomorphism follows only under the separating hypothesis of Theorem 4.4. Without it, the justified conclusion is that the effective operator or connectivity descriptions differ.

We return to the conceptual implications of this distinction in the Discussion.

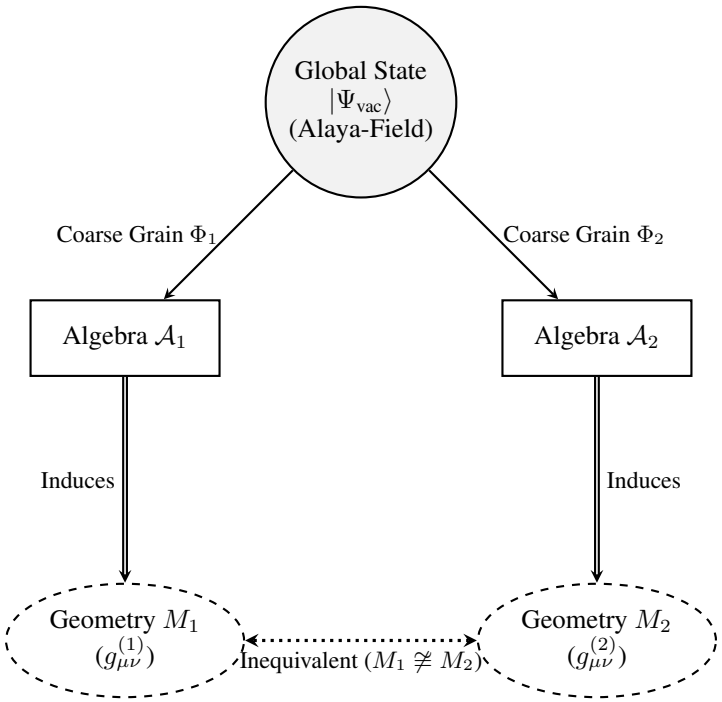



Figure 1: Schematic of the conditional HAFF construction. A single global state is reduced by two coarse-graining channels. The outputs define different effective connectivity data; only a stated reconstruction dictionary promotes them to metrics $g_{\mu\nu}^{(1,2)}$. The dependence is on coarse-graining at the level of effective description, not on an observer in a fundamental sense.

## 5 Discussion

### 5.1 Relation to Existing Frameworks

The framework developed in this work is not intended as a replacement for existing interpretational or algebraic approaches to quantum theory. Rather, it is best understood as orthogonal to several well-established lines of thought, addressing a distinct structural question. In this section, we briefly clarify its relation to three representative frameworks:

#### 5.1.1 Relation to QBism

QBism emphasizes the role of agents and their personal probability assignments in quantum theory, interpreting the quantum state as an expression of subjective belief rather than an objective property of a system. In contrast, the present framework assumes a single, objective global quantum state throughout. No agent-dependent elements enter the formalism, and no interpretational commitments regarding belief, experience, or decision theory are required.

The point of contact lies solely in the rejection of a privileged subsystem decomposition. While QBism attributes this absence to the primacy of the agent, our framework treats it as a structural feature of quantum theory itself. Coarse-grainings are not chosen by agents, but instantiated by concrete physical interaction structures. The resulting multiplicity of effective descriptions reflects physical underdetermination rather than subjectivity.

#### 5.1.2 Relation to Many-Worlds and Decoherence-Based Approaches

Many-Worlds–type interpretations and decoherence-based accounts provide a compelling explanation of classical behavior within quantum mechanics by analyzing branching structures relative to a fixed subsystem decomposition. Our framework does not modify this analysis, nor does it introduce an alternative account of branching or classicality.

The difference lies at a prior level. Decoherence theory presupposes a tensor factorization into system, apparatus, and environment. Here, we instead ask how such subsystem structures arise in the first place. In this sense, the framework is complementary to decoherence-based approaches: it leaves their dynamical conclusions intact while removing subsystem factorization from the list of fundamental assumptions.

#### 5.1.3 Relation to Algebraic Quantum Field Theory

The closest structural affinity of the present work is with algebraic quantum field theory (AQFT). In AQFT, observable algebras are taken as primary, and states are defined as positive linear functionals over these algebras, without reliance on a global tensor product structure. Our use of observable algebras and their representations is directly inspired by this tradition.

The present framework may be viewed as extending this algebraic perspective by emphasizing the role of coarse-graining relations between algebras. Different choices of coarse-graining induce different effective representations and, consequently, different entanglement structures. The novelty lies not in the algebraic formalism itself, but in using it to analyze the emergence and non-uniqueness of geometric descriptions.

Taken together, these comparisons situate the present work as a structural investigation into the conditions under which subsystem structure and geometry emerge. It neither commits to a particular interpretation of quantum mechanics nor proposes a new dynamical law, but instead clarifies how several existing frameworks implicitly rely on assumptions that can be made explicit and, in some cases, relaxed.

### 5.2 Scope and Limitations

The present work is concerned with structural consistency rather than phenomenological prediction. Observable consequences depend on the physical mechanisms implementing a given coarse-graining, which lie beyond the scope of this paper. This is analogous to effective field theory, where multiple UV completions may share the same low-energy structure. Future work will explore concrete physical realizations and their empirical signatures.

### 5.3 Philosophical Implications

We emphasize that the framework assumes a single, objective global quantum state $|\Psi_U\rangle$. What is coarse-graining–dependent is not reality itself, but the effective structures used to describe it. This position is compatible with scientific realism while acknowledging the role of operational context in physical description.

### 5.4 Future Directions

While the present work is deliberately limited to a structural analysis of subsystem emergence and geometry within a fixed global quantum state, it naturally opens several directions for further investigation. We emphasize that the following points are not results established here, but rather indicate possible extensions where the current framework may provide useful conceptual or technical guidance.

#### 5.4.1 Dynamical Models of Coarse-Graining Selection

In this work, coarse-graining structures are treated as fixed relational inputs, instantiated by concrete physical interaction patterns. A natural next step is to investigate whether such coarse-grainings can themselves be characterized dynamically.

One possible direction is to study how interaction Hamiltonians, coupling strengths, or network topologies bias the emergence of particular subalgebra structures over others, in contact with explicit quantum-mereology and locality-selection programmes [16, 17]. This could clarify under what physical conditions certain factorizations become robust or persistent, and whether transitions between inequivalent coarse-grainings admit an effective dynamical description.

Importantly, such an analysis would remain compatible with globally unitary evolution, treating coarse-graining selection as an emergent, effective phenomenon rather than a modification of fundamental dynamics.

#### 5.4.2 Connections to Quantum Information and Complexity

The non-uniqueness of emergent geometry highlighted here suggests a close connection to quantum information–theoretic notions such as entanglement structure, operator complexity, and resource constraints.

Future work could explore whether preferred geometric descriptions correlate with informational criteria—for example, minimal description length, stability under noise, or computational accessibility of observables. Such considerations may help explain why certain coarse-grainings are physically salient, even when many are formally admissible.

This perspective may also provide a bridge to recent work on complexity-based approaches to spacetime emergence, without committing to any particular complexity measure at the present stage.

#### 5.4.3 Extensions to Quantum Field Theory and Continuum Limits

While the framework has been formulated in abstract Hilbert space terms, an important open question concerns its implementation in quantum field–theoretic settings, where issues of locality, algebraic nets, and continuum limits arise.

In particular, it would be valuable to examine how coarse-graining relations between observable algebras interact with the locality structures emphasized in algebraic quantum field theory, and whether familiar spacetime geometries can be recovered as stable fixed points of such relations.

We stress that the present work does not resolve these questions, but provides a structural language in which they can be posed more precisely.

#### 5.4.4 Empirical and Phenomenological Implications

At the level developed here, the framework is primarily structural and conceptual. Nevertheless, future investigations may ask whether different coarse-graining choices lead to distinguishable effective descriptions, for example in semiclassical regimes, quantum gravity–motivated models, or analogue systems.

Such studies could clarify whether the non-uniqueness of emergent geometry has observable consequences, or whether physical constraints effectively suppress this freedom in realistic settings.

Any empirical analysis would necessarily require additional assumptions beyond those adopted in this work, and thus lies outside its present scope.

#### 5.4.5 Conceptual Clarifications and Interpretational Interfaces

Finally, although the framework is intentionally neutral with respect to interpretations of quantum mechanics, it may serve as a useful interface for comparative studies. By making explicit the structural assumptions underlying subsystem decomposition and geometry, it could help clarify which features are interpretation-dependent and which arise more generally from the formalism itself.

We view this not as an attempt to adjudicate between interpretations, but as an opportunity to sharpen the questions they address.

Taken together, these directions suggest that the framework developed here is best viewed as a scaffold: it does not dictate specific physical models, but provides a structured setting in which questions about subsystems, geometry, and emergence can be formulated with greater precision.

## 6 Conclusion

We have presented a framework in which subsystem structure is not presupposed, but is modelled through coarse-graining over selected observable algebras. The central results are:

1. For a finite, physically preselected family of factorizations, Haar-typical states are highly entangled across every member under the stated dimension hypotheses (Theorem 2.1); no claim is made over all possible factorizations or about a preferred factorization.
2. For a fixed pair of distinct finite-dimensional channels, their outputs are generically separated; a generic entropy difference requires the explicit non-degeneracy hypothesis in Theorem 3.5.

3. When a separating reconstruction dictionary is supplied and the compared entropy profiles are inequivalent in its declared sense, the reconstructed geometric data are not related by a diffeomorphism (Theorem 4.4).

The framework is deliberately limited in scope. We have not proposed new dynamics, derived empirical predictions, or resolved interpretational debates. Instead, we have examined the structural consequences of removing subsystem factorization from the list of fundamental assumptions.

Within the stated channel and reconstruction hypotheses, this investigation shows how geometric data can depend on the chosen coarse-graining context. It does not establish a geometry for an arbitrary quantum state or coarse-graining without an additional reconstruction dictionary. This perspective is compatible with, and complementary to, existing approaches including decoherence theory, holographic duality, and algebraic quantum field theory.

Several open questions remain. Can coarse-graining structures themselves be characterized dynamically? Do informational or complexity-based criteria select physically preferred coarse-grainings? Can the framework be extended to quantum field theory and reconciled with standard locality structures? These questions lie beyond the present scope, but the structural setting developed here provides a language in which they can be formulated with greater precision.

Ultimately, this work suggests that the relationship between quantum states, subsystems, and geometry is more subtle than the standard picture implies. By making explicit an assumption that is often left implicit, we hope to have clarified the conditions under which emergence occurs and opened new avenues for investigating the foundations of quantum theory and spacetime.

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